Hot Air Oven



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Hot Air Oven

What it is used for

To heat and dry laboratory materials or perform approved thermal procedures using a hot air oven. To maintain the required temperature and exposure time while protecting samples and users from heat hazards.

Main parts

- Insulated oven chamber
- Heating elements
- Temperature controller and sensor
- Shelves or trays
- Door and heat-resistant seal
- Ventilation openings and control panel

Before you start

1. Read the oven manual and confirm that the intended material is suitable for hot-air heating.
2. Inspect the chamber, door seal, shelves, power cable, plug, and controls for damage.
3. Ensure the oven is clean and free of spilled chemicals or combustible residue.
4. Confirm that the temperature range and timer settings suit the approved procedure.
5. Use suitable heat-resistant containers and arrange samples so that air can circulate.
6. Ensure ventilation is adequate and the oven is placed away from flammable materials.

How to use

1. Place the oven on a stable, heat-resistant surface with sufficient clearance around it.
2. Switch off the oven before loading samples or adjusting shelf positions.
3. Load samples evenly without blocking ventilation openings or touching heating elements.
4. Close the door securely and set the required temperature and time.
5. Switch on the oven and allow it to reach the set temperature.
6. Monitor the displayed temperature and use an approved independent thermometer if the procedure requires verification.
7. Keep the door closed during heating unless the procedure specifically requires access.
8. When the cycle is complete, switch off the heater and allow the chamber to cool as required.
9. Use heat-resistant gloves and suitable tools to remove hot trays and containers.
10. Place hot items on a heat-resistant surface and label or handle them according to the procedure.
11. Switch off and disconnect the oven before cleaning or maintenance when safe to do so.
12. Clean the chamber only after it has cooled and follow the manufacturer's cleaning instructions.

⚠ Safety

- Do not heat flammable, explosive, volatile, or chemically incompatible materials unless the oven is specifically designed and approved for that use.
- Never use a hot air oven for materials that require a specialized sterilization or controlled-atmosphere system unless approved for that purpose.
- Avoid overloading the chamber and keep samples away from heating elements.
- Wear suitable heat-resistant gloves, eye protection, and laboratory clothing when handling hot items.
- Do not touch the chamber or trays with bare hands while hot.
- If smoke, unusual odour, or overheating occurs, switch off the unit if safe, keep clear, and follow laboratory emergency procedures.

Care & storage

- Clean the chamber and shelves after cooling using approved materials.
- Inspect the door seal, shelves, and electrical cord regularly.
- Keep air vents clear and avoid blocking the temperature sensor.
- Check temperature accuracy according to the laboratory schedule.
- Record faults and arrange qualified maintenance when needed.

Common mistakes to avoid

- Setting the wrong temperature or exposure time.
- Overloading the oven and restricting air circulation.
- Heating unsuitable flammable or volatile substances.
- Removing hot trays without protective equipment.
- Cleaning the oven while it is hot or connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.